

NHAM HOANG DUC

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SUMMARY

Data engineer at Viettel Telecom, where I rebuilt the pipelines behind a company-wide fee settlement system. At Aubot, I took a multi-agent path finding planner from recent literature into production on a live AGV fleet. Building a hybrid retrieval system over internal policy documents was where the two halves of my work met: the model was only as good as the pipeline feeding it. I want to keep working at that intersection: the data infrastructure that makes large models work, from the pipelines that train them to the systems that serve them.

EDUCATION

FPT University

Hanoi, Vietnam

Bachelor of Information Technology | GPA: 8.08/10

2022 – August 2026

- Specialization: Software Engineering. Graduation Capstone Project: SWES - Smart Warehouse Execution System (Multi-AGV Orchestration). **Grade: 9.0/10 (A+)**.
- Relevant courses: Statistics and Probability; Discrete Mathematics; Data Structures and Algorithms; Introduction to Databases; Operating Systems; Computer Organization and Architecture; Computer Networking; Internet of Things.

WORK EXPERIENCE

Viettel Telecom

Hanoi, Vietnam

Data Engineer (Full-time)

July 2025 – now

- Sales Fee System v3 could only settle fees **monthly**: data left the Data Lake as **TXT files** and was enriched and aggregated inside Oracle.
- Rebuilt the pipeline on the **Data Lakehouse**, with **Apache Spark** handling aggregation, enrichment and fee calculation across terabytes of data. Pipeline runtime fell from **over 30 minutes to under 8 minutes**, and settlement moved from monthly to **daily**.
- Migrated **17 tables** from v3 with Apache Hop and consolidated external sources through Apache NiFi, with data integrity preserved across the cutover.
- Deployed a **phone number masking system** in partnership with China-ASEAN Information Harbor Co., Ltd., configuring the KrakenD API Gateway and streaming logs into the **ELK stack** through Fluentd.

Viettel Group

Hanoi, Vietnam

Data Engineering Intern, Viettel Digital Talent Program (DS/AI Track)

April 2025 – June 2025

- Built ETL workflows following the **Medallion architecture** (Bronze/Silver/Gold) on Databricks, using Spark SQL to transform data across each layer.
- Researched and applied fine-grained access control with Apache Ranger, Databricks Unity Catalog and lakeFS to strengthen lakehouse governance.
- Developed a hybrid RAG system that answers staff questions on which company-issued policy applies, returning the governing policy document with traceable citations.

PROJECTS

SWES - Smart Warehouse Execution System (Multi-AGV Orchestration)

June 2026 – August 2026

Research and Development, Multi-Agent Path Finding | Aubot (CF Group) | Graduation capstone project

- openTCS ships with a **Dijkstra** router that plans each vehicle independently, which left the AGVs deadlocking against one another in regular operation.
- Built a joint multi-agent planner in the openTCS Java kernel, evolving from **PIBT to LaCAM and LaCAM***, extended to continuous order arrival with LLLG local guidance (SoCS-26) and an **Action Dependency Graph** against corridor deadlock.

- Ported these undirected-graph algorithms onto a **directed industrial roadmap** with vehicle footprints and headings, and replaced greedy assignment with **global min-cost matching** (Hungarian algorithm) on road-graph distances.
- Deployed on Aubot's **live AGV fleet**.

Hybrid RAG System | Viettel Digital Talent Program

April 2025 – June 2025

Python, FastAPI, Streamlit, Milvus, MinIO, Docker, BGE-M3, Gemma 3

- Built a hybrid retrieval system that answers staff questions on which company-issued policy applies, returning the governing policy document with traceable citations.
- Indexed documents with **BGE-M3 embeddings** (1024-dim) in Milvus and combined dense vector search with BM25 sparse search, improving recall and ranking quality.
- Integrated Gemma 3 models as the generative backbone and benchmarked performance across model sizes.
- Containerized all services (FastAPI, Streamlit, Milvus, MinIO, Etc) with Docker Compose and health monitoring.

NYC Taxi Data Pipeline

Terraform, Apache Airflow, BigQuery, dbt, Apache Spark, Apache Kafka, Apache Flink

- Built an automated pipeline for NYC taxi trip data on Google Cloud, provisioning all resources with **Terraform** for reproducibility.
- Orchestrated ingestion and transformation with an **Apache Airflow DAG**, loading results into BigQuery and modeling them with dbt.
- Processed large-scale trip data with Apache Spark and implemented **real-time streaming** with Apache Kafka and Apache Flink.

AWARDS & HONORS

Viettel Digital Talent Program | Data Engineer (DS/AI Track)

2025

- Selected in the **top 3.6%** of **5,700+ applicants** nationwide, advanced to the full-time phase, and offered full-time employment while still in the third year of university.

Honorable Student of the Semester | FPT University | Fall 2023, Spring 2024 and Summer 2024

SKILLS

Languages: Vietnamese (native), English (IELTS 6.5)

Programming: Python, Java, TypeScript, SQL

Data engineering: Apache Spark, Hadoop, Kafka, Flink, Airflow, NiFi, Apache Hop, dbt, Databricks

Databases and cloud: PostgreSQL, Oracle, MinIO, Google Cloud, Docker

Backend and web: Spring Boot, NestJS, .NET, FastAPI, Flask, React

Monitoring and reporting: ELK Stack, Fluentd, KrakenD, Power BI